

Binary Search on Decimals & ~~Problem Solving~~

(Next class)

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Problem 1: Sqrt(x)

Given a +ve integer x find out the highest integer y such that $y^2 \leq x$

$$x = \underline{\underline{37}}$$

$$y \rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, \dots\}$$

↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑

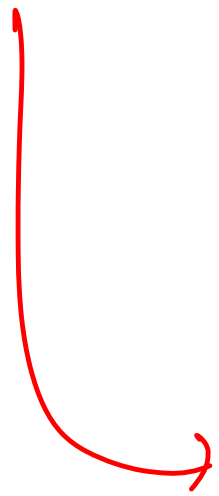
✓ ✓ ✓ ✓ ✓ ✓ ✗ ✗

$f(y) \rightarrow$ T if $y^2 \leq x$ and f o/w

T T T T T T T f f f f f f f f

Search Space $\rightarrow$ given input is X

$$(0 \leq X \leq 10^9)$$



$$(0 \text{ to } 10^5)$$

$$\log_2(10^5) \leq \underline{\underline{19}}$$

$$(0 \text{ to } 10^9)$$

$$\log_2(10^9)$$

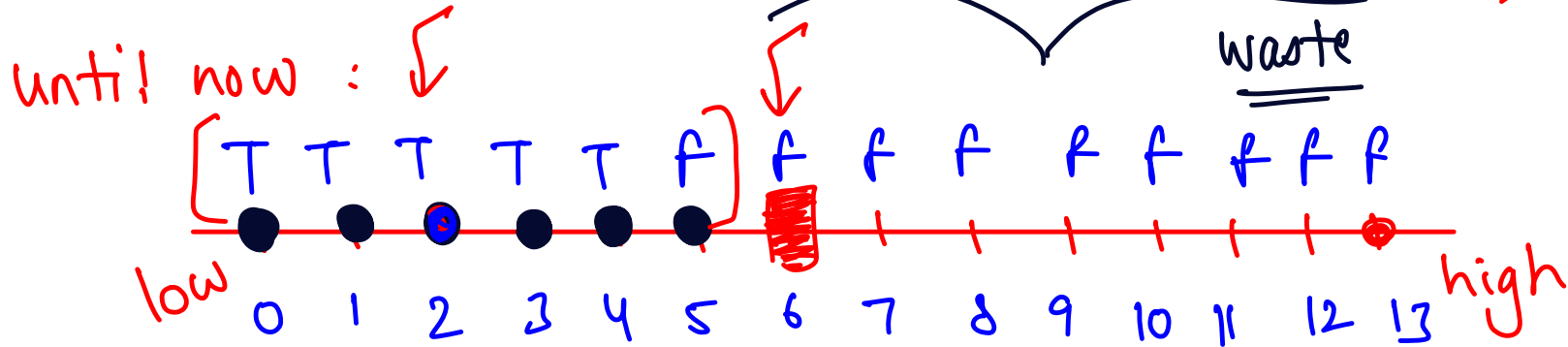
$$\leq \underline{\underline{30}}$$

Binary Search on Decimals

Find the highest real number between L and R which returns a true on a monotonic predicate function (TTTTTTTTTFFFFF).

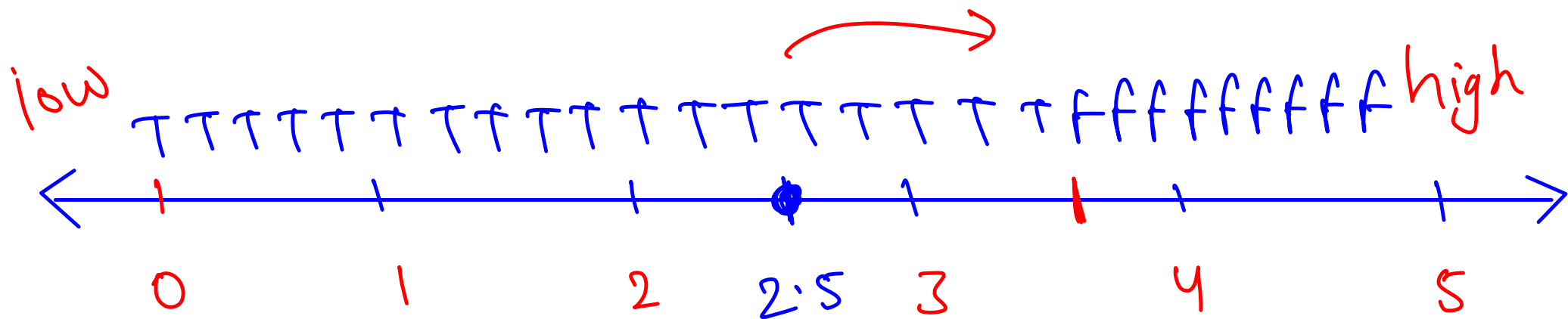
- Let's solve the same Sqrt(x) problem but on real space now.
- Do we have finite candidate answers?

No



$$\text{low} = \text{mid} + 1$$

$$\text{high} = \text{mid} - 1$$



$$low = mid + 1$$

$$= mid + 0.5$$

$$= mid + 0.001$$

$$= mid + 0.0000000002$$

$$= mid + 0.00000000000000000001$$

a, b

$$\frac{a+b}{2}$$

$$10/3 \longrightarrow 3.33333333 \dots$$

⇒

we want ans
to be correct
upto 5 decimal
places

$$3.33333 \text{ — } \checkmark$$

$$3.33333\underline{9} \text{ — } \checkmark$$

$$3.333323 \text{ — } \times$$

Correct upto X decimal
places 5%

Real $\rightarrow$ 15.00012349

Output $\rightarrow$ 15.00012350

15.00011250 X

X = 5

Output must have
relative error $\leq$ y
 $y = 0.00001$ 95%

Real ans = 1.0027298

output = 1.0027300

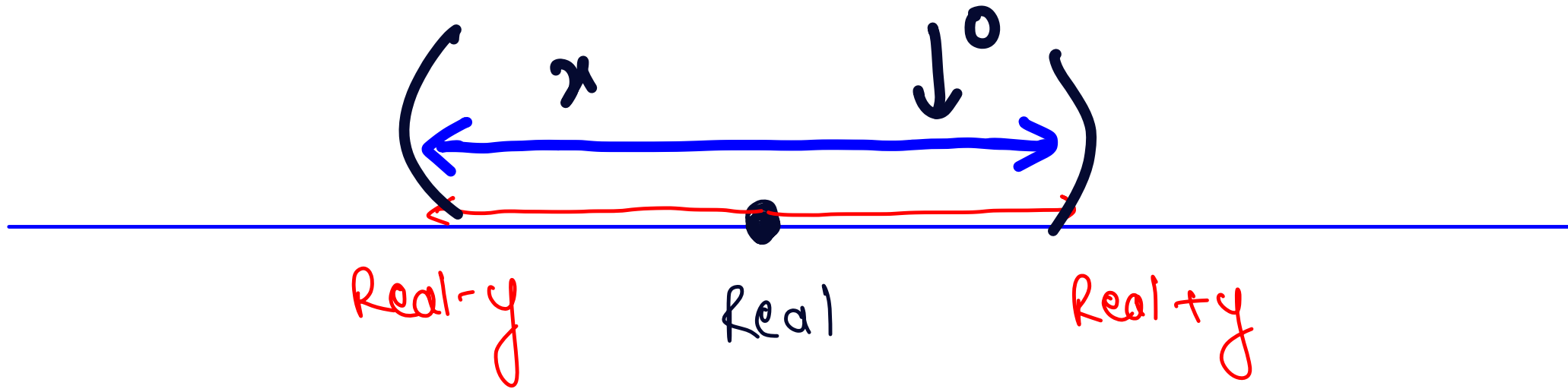
$$|\text{Real} - \text{output}| \leq y$$

Relative error being less than 10^{-6} $10/3$

Real Ans = $3.333333333333 \dots$

Output = 3.33333350

$$| \text{Real} - \text{output} | \leq 10^{-6}$$



$$(\text{Real} - x) \leq y$$

$$\text{Rel. error} \leq y$$

Examples

$$y = 10^{-2} = \underline{\underline{0.01}} \quad | \quad \begin{array}{l} \text{Real} = 0.012 \\ \text{Output} = 0.02 \end{array}$$

$$| \text{Real} - \text{Output} | \leq y$$

$$| 0.012 - 0.020 | \leq 0.01$$

$$0.008 \leq 0.01$$

If relative error requirement is for $\gamma = 10^{-3}$
then if we print any answer which is
correct upto 4 decimal places will work

$$| \text{Real} - \text{output} | \leq 0.001$$

$$\text{Real} = 1.0023\textcolor{red}{4}$$

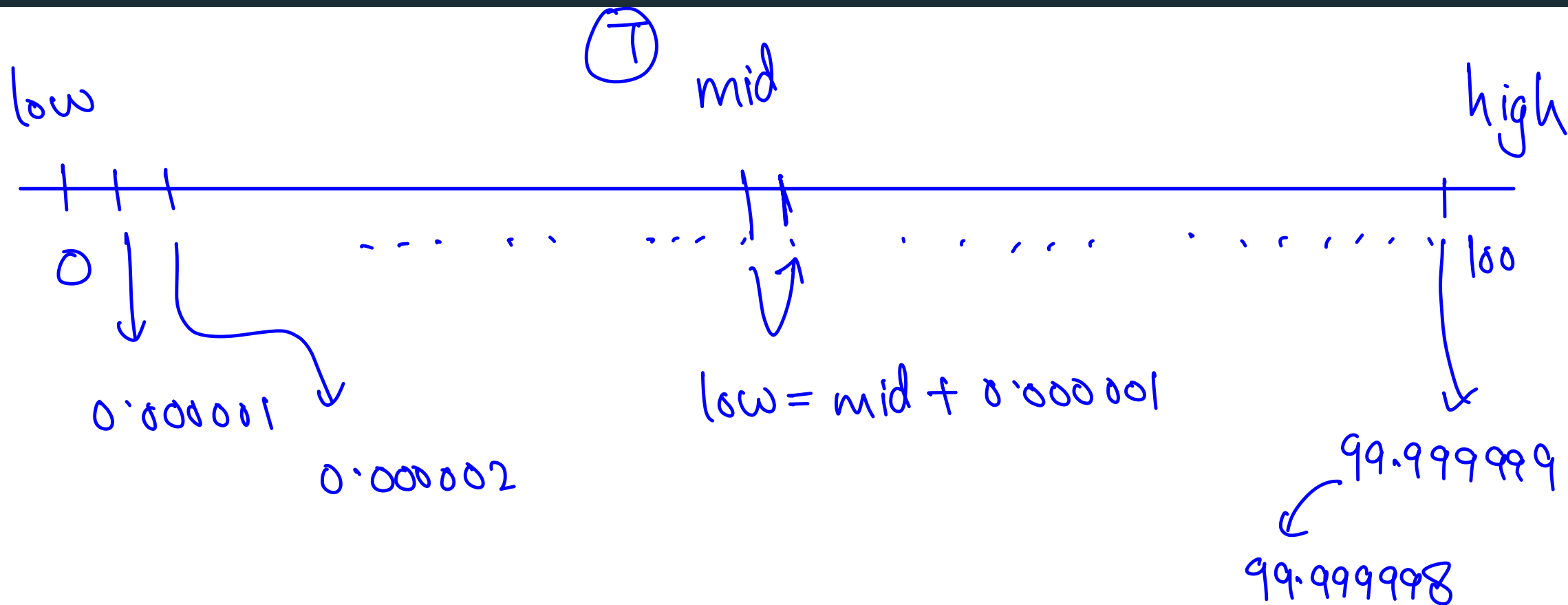
$$\text{Output} = 1.0022\textcolor{red}{9872}$$

$$\text{diff} \leq \textcolor{red}{\underline{\underline{0.0001252}}}$$

How to handle infinite candidates?

We can make the number of candidates finite by fixing the number of decimal places in the answer.

Example: Find the highest value k such that $k^2 \leq x$ the answer is correct upto 6 decimals places



How many iterations?

Without any precision requirement

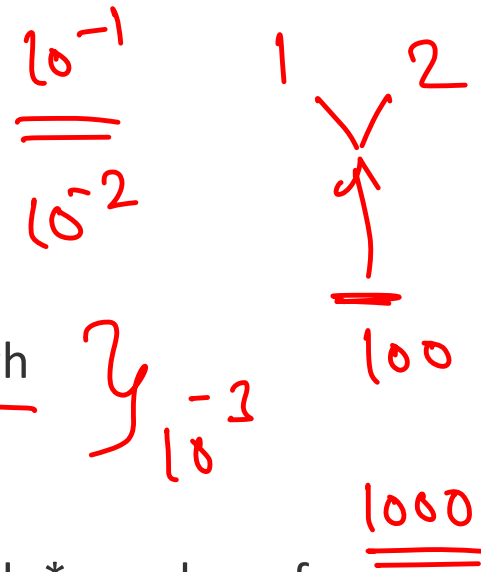
- Search space = number of integers from low to high
- Iterations = $\log_2(\text{high} - \text{low})$

With precision (k decimal places)

- Search space = number of integers from low to high * number of decimal numbers between 2 integers with the given precision
- Iterations = $\log_2((\text{high} - \text{low}) * 10^k)$

Example: low = 0, high = 10^{18} and precision = 6 decimal places

Iterations = $\log_2(10^{18} * 10^6) = 85$ (approx)



How to make the code always stop?

- Find out the number of iterations before writing the code.
- Write a for loop for that many iterations + some extra iterations.
 - No need to handle any low <= high type of conditions.

Some nice results to know while calculating number of iterations

- $\log_2(1000) = 10$
- $\log_a(b) + \log_a(c) = \log_a(b * c)$
- $\log_2(1000 * 1000) = \log_2(1000) + \log_2(1000) = 10 + 10$
- $\log_2(1000^k) = k * 10$

$$2^{10} = 1024$$

$$\log_2(1000)$$



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Double has some precision $\rightarrow$ 16 decimal places



long double $\rightarrow$ 2x the precision of double